🐋 Orcas & the Salish Sea: Reading · ANSWER KEY

Biology A · Unit 1 · Level 3 (closer to the class work) · goes with Unit 1 Orca: A1 J32, A2 webquest, A3 food web, A4 Trace the Toxin, A5 PCB articles, A6 Ocean noise, A9 Orca Test

Name: _____

Why Are the Southern Residents in Trouble?

The Southern Resident killer whales (SRKW) live in the Salish Sea, the waters between Washington and British Columbia. They travel in three pods, J, K and L, and there are fewer than 80 of them. Orca families are close: a mother and her young form a matriline, related families form a pod, and pods that share calls form a clan.

Southern Residents eat mostly Chinook salmon. That puts them at the top of a long food web. Phytoplankton are producers: they make food from sunlight. Zooplankton eat phytoplankton, herring eat zooplankton, salmon eat herring, and orcas eat salmon. The arrows in a food web show energy moving from one living thing to the next.

PCBs are toxic chemicals that wash into the sea from old industry and storm drains. PCBs are fat-soluble, so they are stored in fat, like orca blubber. Each animal eats many animals from the level below, so the toxin becomes more concentrated at every step. This is called biomagnification, and it means orcas carry the most.

Orcas hunt with echolocation: they make clicks and listen for the echoes that bounce off fish. Boat and ship noise covers those echoes, so hunting takes more time and energy. When orcas cannot find enough food, they burn their blubber, and the stored PCBs are released into their blood.

In 2018 the orca J32 died while carrying a calf. Scientists found that lack of food, toxins and noise worked together. Solutions include restoring salmon rivers (like removing the Elwha dams), slowing ships, keeping toxins out of storm drains, and planting trees to shade salmon streams.

Check your reading

1. What do Southern Resident orcas mostly eat?
Chinook salmon.

2. Explain biomagnification in your own words.
Toxins get more concentrated at each step up the food chain, so top predators like orcas have the most.

3. Name the three threats that worked together in J32's death.
Lack of food (salmon), toxins (PCBs), and noise.